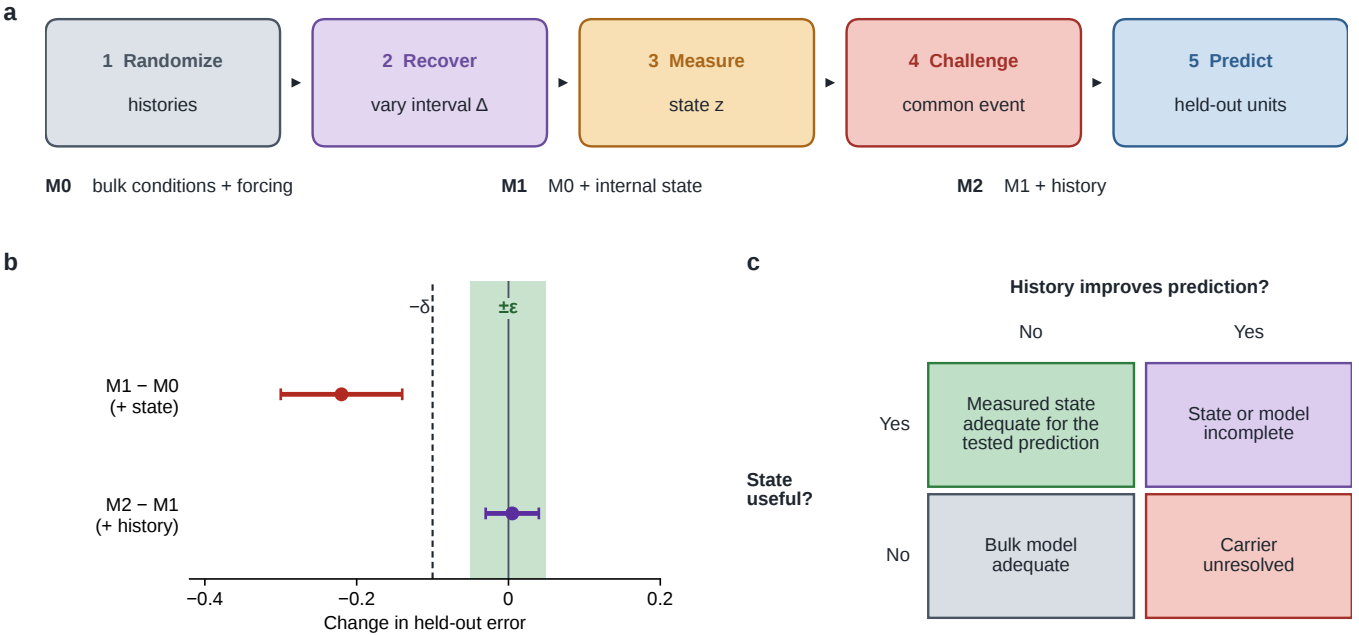
a
1 Randomize
histories
2 Recover
vary interval Δ
3 Measure
state z
4 Challenge
common event
5 Predict
held-out units
M0 bulk conditions + forcing
M1 M0 + internal state
M2 M1 + history
b
−δ
±ε
M1 − M0
(+ state)
M2 − M1
(+ history)
−0.4
−0.2
0
0.2
Change in held-out error
c
History improves prediction?
No
Yes
State useful?
Yes
No
Measured state adequate for the tested prediction
State or model incomplete
Bulk model adequate
Carrier unresolved